



Microsoft Power Automate for End Users

Two-day course book

Collect, report, remind, approve and release: automate a training coordinator's routine with automated, instant and scheduled cloud flows, Copilot and solutions.

Microsoft Power Automate for End Users

Two-day course book · Participant edition, version 1.1

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Online resources: github.com/Asraf-JS/Microsoft-Power-Automate

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Screenshots show the product at the time of writing. Microsoft updates its apps often, so labels and layouts in your tenant may differ slightly.

About the author

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He runs his training practice solo, which means he has personally felt every inefficiency this book tries to fix.

Contents

	Before you begin	8
1	Understand Power Automate	11
	What You Will Learn	11
	Your Business Scenario	11
	1.1 What Is Automation?	11
	1.2 Recognise the Parts of a Flow	12
	1.3 Choose How the Flow Starts	12
	1.4 Connectors and Connections	13
	1.5 Explore Power Automate	13
	1.6 Plan Before You Build	14
	Independent Practice: Plan a Weekly Reminder	14
	Tips and Troubleshooting	15
	Lesson Summary	15
	Hands-on exercises	16
2	Prepare Your Training Workspace	17
	What You Will Learn	17
	2.1 Create the Training Folders in OneDrive	17
	2.2 Create the Excel Workbook	18
	2.3 Prepare a Test Email	18
	2.4 Confirm Forms Readiness	18
	2.5 Confirm the Teams Training Destination	19
	2.6 Confirm the Power Automate Environment	19
	Independent Practice	19
	Lesson Summary	19
	Hands-on exercises	20
3	Save Email Attachments Automatically	21
	What You Will Learn	21
	Before You Begin	21
	3.1 Create the Automated Cloud Flow	22
	3.2 Configure the Trigger	22
	3.3 Add Apply to Each	22
	3.4 Add the PDF Condition	22
	3.5 Create the OneDrive File	23
	3.6 Test a Positive Case	23
	3.7 Test a Negative Case	23
	3.8 Decide What Happens to Duplicate Names	24
	Independent Practice	24
	Tips and Troubleshooting	24

Lesson Summary	24
Hands-on exercises	25
4 Create and Distribute an Excel Summary	26
What You Will Learn	26
How the Flow Works	26
Before You Begin	26
4.1 Confirm the Source Table	27
4.2 Create an Instant Cloud Flow	27
4.3 Read the Excel Records	27
4.4 Choose the Report Columns	28
4.5 Build One HTML Table	28
4.6 Send the Coordinator Summary	28
4.7 Test and Inspect	29
4.8 Optional Formatting Practice	29
4.9 Optional Teams Update	29
Independent Practice	29
Troubleshooting	29
Lesson Summary	30
Hands-on exercises	31
5 Send Weekly Training Reminders	32
What You Will Learn	32
How the Flow Works	32
Before You Begin	32
5.1 Confirm the Training Register	33
5.2 Create the Weekly Schedule	33
5.3 Produce a Readable Local Timestamp	33
5.4 Read the Excel Rows	34
5.5 Process One Row at a Time	34
5.6 Prepare the Reminder Email	34
5.7 Save and Check the Flow	35
Trainer-Controlled Test	35
Independent Practice	35
Troubleshooting	35
Lesson Summary	36
Hands-on exercises	37
6 Process a Training Approval	38
What You Will Learn	38
How the Flow Works	38
Before You Begin	38
6.1 Prepare the Request Form	39
6.2 Create the Automated Flow	39

6.3 Select the Form	39
6.4 Retrieve the Answers	39
6.5 Request the Decision	39
6.6 Branch on the Outcome	40
6.7 Prepare the Approved Message	40
6.8 Prepare the Rejected Message	40
6.9 Save and Check the Flow	40
Trainer-Controlled Test	41
Independent Practice	41
Troubleshooting	41
Lesson Summary	41
Hands-on exercises	42
7 Capstone: Weekly Reminders and Summary	43
What You Will Learn	43
How the Flow Works	43
Before You Begin	43
7.1 Make a Safe Capstone Copy	44
7.2 Return Excel Dates in a Consistent Format	44
7.3 Select Upcoming Incomplete Records	44
7.4 Start the Reminder Counter	44
7.5 Loop Through the Filtered Rows	44
7.6 Count Each Prepared Reminder	45
7.7 Review the Learner Reminder	45
7.8 Decide Whether a Summary Is Needed	45
7.9 Prepare the Coordinator Summary	45
7.10 Handle a Week with No Matching Rows	45
7.11 Save and Check the Capstone	46
Trainer-Controlled Test	46
Independent Practice	46
Troubleshooting	46
Lesson Summary	47
Hands-on exercises	48
8 Plan an Automation with Copilot	50
What You Will Learn	50
The Business Problem	50
Before You Begin	50
8.1 Open the Copilot Planning Experience	51
8.2 Describe the Complete Business Outcome	51
8.3 Review the First Suggestion	51
8.4 Clarify the Filter and Count	51
8.5 Identify an Unnecessary Action	51
8.6 Request a Streamlined Design	52

8.7 Validate the Final Suggestion	52
8.8 Exit Without Creating the Flow	52
Independent Practice	52
Troubleshooting	52
Lesson Summary	53
Hands-on exercises	54
9 Move Automations with Environments and Solutions	55
What You Will Learn	55
The Migration Path	55
Before You Begin	55
Environments in Plain Language	56
9.1 Confirm the Current Environment	56
9.2 Create the Solution	56
9.3 to 9.7 Add the Capstone Flow	57
9.8 Verify the Solution Components	57
9.9 Review the Solution Before Export	57
9.10 to 9.12 Publish and Export	57
9.13 and 9.14 Switch Environment and Start the Import	58
9.15 Repair Connections and Settings	58
9.16 Verify and Test Safely	58
Independent Practice	58
Troubleshooting	58
Lesson Summary	59
Hands-on exercises	60

Before you begin

How to use this book

This book collects the notes and copy-paste values from the two-day Microsoft Power Automate for End Users course in one place, so you can follow along offline and keep it at your desk afterwards.

Each chapter matches one chapter of the course and has two parts:

- **Notes** explain the concepts and walk you through each build step, with checkpoints and troubleshooting.
- **Hands-on exercises** give you every flow name, expression and text value you type, ready to copy.

For screenshots of each step, use the participant manual PDF that comes with the course. This book follows the same section numbers, so section 3.4 here is section 3.4 there.

Look out for the coloured boxes. Blue boxes are notes, teal boxes are tips, purple boxes are things to try, and orange boxes are key points and warnings.

Prefer to copy expressions rather than retype them? Everything in this book is also online at github.com/Asraf-JS/Microsoft-Power-Automate, where each grey box has a copy button. Copying from there is safer than copying from a PDF, which can add line breaks or change quotation marks.

Course scenario

Throughout the course you automate the work of one training coordinator.

You coordinate training for a small organisation. Course materials arrive by email, session details live in an Excel register, requests need approval, and participants need reminders.

Each chapter turns one of those manual tasks into a flow:

Stage	What happens
Plan	Learn how flows work and write a plan before building anything
Prepare	Set up the OneDrive folders, Excel register, test email, Forms and Teams destinations
Collect	Save PDF attachments from training emails to OneDrive automatically
Report	Send the coordinator a one-click summary table from the Excel register
Remind	Send weekly reminders to every participant on a schedule
Approve	Route a training request form through an approval and post the result to Teams
Refine	Filter reminders to upcoming incomplete sessions, count them and summarise the week

Design	Use Copilot to plan a flow, then review and correct its suggestion
Release	Package the finished flow in a solution and export it for another environment

Ground rules

Important: Use only fictional data. Keep flows turned off until your trainer asks you to test them, and only send test emails to addresses your trainer has approved. Testing a flow can create files, send messages, start approvals or change shared data.

Sample files

Four practice files come with the course. All of them are fictional.

File	Used in	What it is
TrainingRegister.xlsx	Chapters 2 to 7	A ready-made register with the tblTraining table and six records. Use it only if your trainer says so
CourseOutline.pdf	Chapters 2 and 3	The attachment the Chapter 3 flow should save
TrainerPhoto.jpg	Chapters 2 and 3	The attachment the Chapter 3 flow should skip
SessionNotes.docx	Chapter 3	Used in the independent practice

Download them from the course repository: github.com/Asraf-JS/Microsoft-Power-Automate.

Your Power Automate training journey

One coordinator's routine, start to finish: turn each repeated manual task into a cloud flow.

FOUNDATIONS

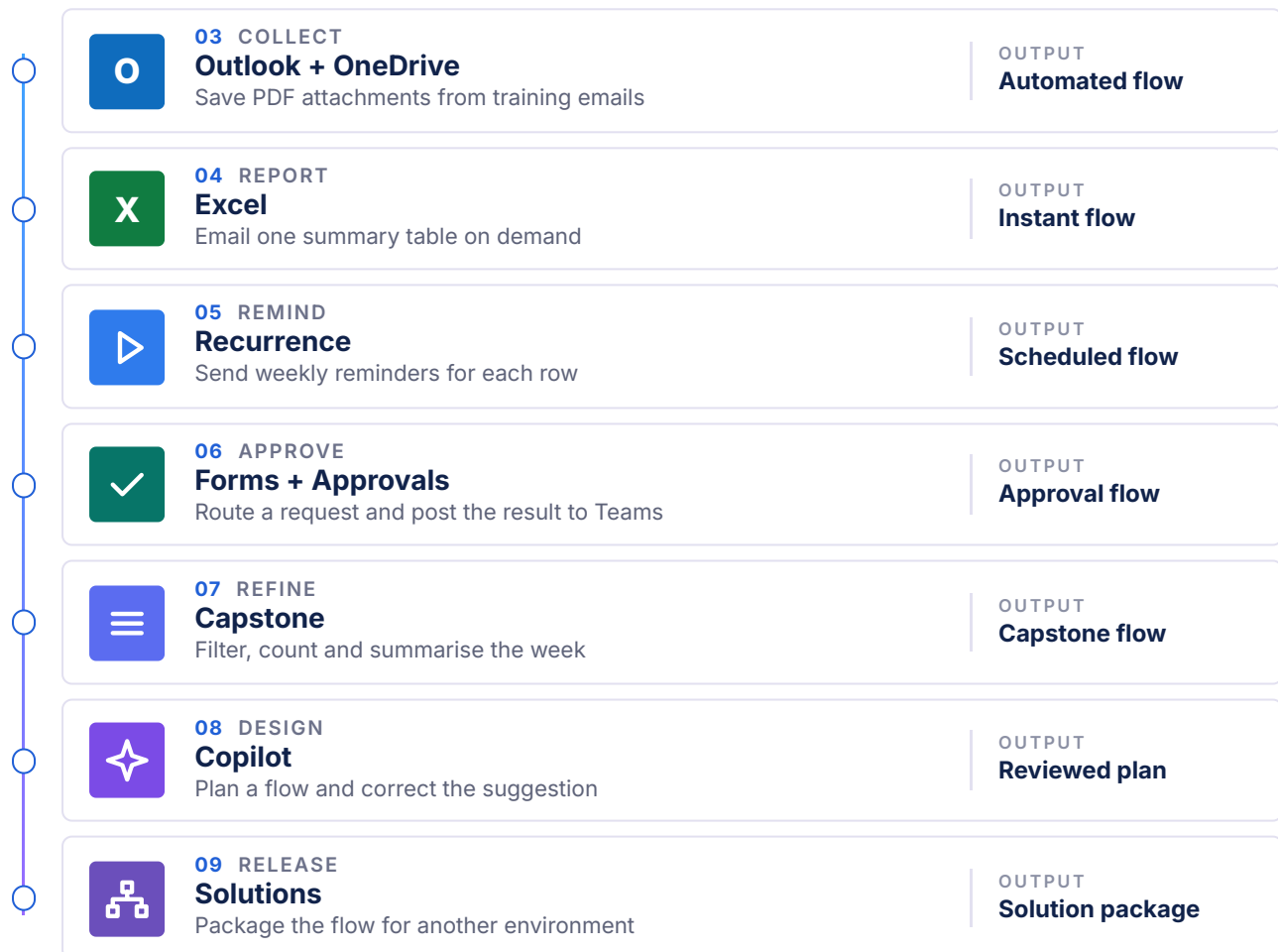
01**Understand Power Automate**

Triggers, actions, flow types and planning

02**Prepare your workspace**

OneDrive folders, the Excel register and test data

THE PROJECT, APP BY APP



Collect → Report → Remind → Approve → Refine → Design → Release

01

CHAPTER 1

Understand Power Automate

Before you build anything, learn the vocabulary of a flow, the three ways a cloud flow can start, and where the main pages live in Power Automate.

Copy-paste values: the planning template for this chapter is on the copy-paste page.

Estimated time: 45 minutes

Your result: A simple automation plan and familiarity with the Power Automate Home and Create pages.

What You Will Learn

- Identify a repetitive business task that could benefit from automation
- Explain the trigger, actions, data, condition and result of a flow
- Choose between an automated, instant and scheduled cloud flow
- Open Power Automate and find Home, Create and My flows
- Write a short plan before building an automation

You need a Windows computer with Microsoft Edge, an internet connection and your training work or school account. You don't need the Excel workbook, a form or any test files yet. You prepare those in Chapter 2.

Your Business Scenario

You coordinate training for a small organisation. Course materials arrive by email, session details are stored in Excel, and participants need reminders. You keep downloading attachments, copying information into messages and checking whether requests were approved.

Each of those tasks has a recognisable starting point and repeatable steps. Over the two days you use Power Automate to do some of those steps for you.

1.1 What Is Automation?

Automation means setting up a repeatable process so software performs the steps when a defined event happens, or when you ask it to run.

Power Automate calls an automation a **flow**. This course builds **cloud flows**, which work with online services such as Outlook and OneDrive for Business.

A flow applies your rule every time, including a wrong one. If you pick the wrong folder or recipient, the flow repeats that mistake. So every flow needs a clear rule and a test.

Tip: Start with a small task whose result you can check easily. A good first flow saves one test attachment to the correct folder.

1.2 Recognise the Parts of a Flow

Take this requirement: *When an email with the subject marker [PA TRAINING] arrives, examine each attachment. If the attachment is a PDF, save it to the training attachments folder.*

Part	Plain-language meaning	Example
Trigger	What starts the flow	A matching email arrives
Action	A step the flow carries out	Create a file in OneDrive for Business
Data	Information a step receives or produces	The attachment's name and contents
Condition	A question that selects a path	Is this attachment a PDF?
Loop	Steps repeated for each item	Check each attachment in the email
Result	The business outcome you verify	Matching files appear in the intended folder

The basic pattern is: something starts the flow, the flow follows its steps, you check the result. A flow only needs a condition or a loop when the process calls for one.

Try it: Imagine a training email with two attachments, `CourseOutline.pdf` and `TrainerPhoto.jpg`. Walk the rule through each one. The PDF is saved, the image is skipped, and the folder should end up with `CourseOutline.pdf` only. If an email has no PDFs at all, the flow still runs successfully and creates nothing.

1.3 Choose How the Flow Starts

Cloud-flow type	When it starts	Training example
Automated	When a specified event occurs	Save attachments when a training email arrives
Instant	When you deliberately start it	Create an Excel summary when you're ready to share it
Scheduled	At a time and frequency you configure	Send training reminders every Monday morning

Pick the type by asking *what should make this process begin?* Sending an email doesn't tell you the type. All three can send email. Only their starting points differ.

Quick check. Pick a type for each request, then check the answers.

1. Whenever a new training request is submitted, ask someone to approve it.
2. At 9:00 AM every Friday, prepare a weekly reminder.
3. Let me generate a summary whenever I need one.

Answers: 1 is Automated, 2 is Scheduled, 3 is Instant.

1.4 Connectors and Connections

A **connector** gives Power Automate a set of triggers and actions for one service. Office 365 Outlook provides email operations. OneDrive for Business provides file operations.

A **connection** links those operations to an account. Whenever a later lesson asks you to sign in to a connector, check that you're using the training account, and that it has permission for the mailbox, folder or resource involved.

The core exercises use standard connectors only. Power Automate Free can create and run cloud flows with standard connectors but can't share flows, so you build and own your own flows. See [Power Automate license types](#).

Tip: Office 365 Outlook and Outlook.com are different connectors. Go by the connector name in the lesson, not the icon.

1.5 Explore Power Automate

The manual has a screenshot for each step (Figures 1.1 to 1.12).

Open the training workspace

1. In Edge, go to make.powerautomate.com and sign in with your training account.
2. Check that the page says **Power Automate** at the upper left.
3. Find the environment name near the top of the page, under **Environments**. An environment is a workspace that holds Power Platform resources, including flows.
4. Compare it with the environment your trainer gave you. If they differ, ask before you build anything.

Find the three cloud-flow choices

1. Expand the left navigation if it's collapsed, then select **Create**.
2. Under **Start from blank** you see five tiles. Three of them create cloud flows, and those are what this course uses:

Tile	Description on the tile	Used in this course
Automated cloud flow	Triggered by a designated event	Yes
Instant cloud flow	Triggered manually as needed	Yes

Scheduled cloud flow	You choose when and how often it runs	Yes
Desktop flow	Automates processes on your desktop environment	No
Process mining	Evaluate and optimize your existing processes and tasks	No

Note: The heading **Three ways to make a flow** refers to the three sections of the page: Start from blank, Start from a template and Start from a connector. Those aren't the three cloud-flow types. The types are three of the five tiles inside the first section.

3. Decide which tile you'd use for a flow that starts when someone submits a form. The answer is **Automated cloud flow**: a form submission is an event that happens without you starting it. Instant means *you* start the flow.

Locate your flows

1. Find **My flows** in the left navigation. You'll come back here to find every flow you build.
2. Select **Home** to return to the start page.

1.6 Plan Before You Build

A short plan makes the designer much easier to use. Here's the plan for the attachment scenario:

Planning question	Example answer
What task repeats?	Saving PDF training attachments from email
What starts it?	An email with the training subject marker arrives
What information is needed?	Attachment name and contents
What rule applies?	Save PDF attachments; skip other file types
Where should the output go?	The training attachments folder in OneDrive for Business
How will I test it?	Use an email with a PDF and an image, then check the folder

Chapter 2 prepares the resources this plan refers to. Chapter 3 turns it into a working flow.

Independent Practice: Plan a Weekly Reminder

Allow five minutes. Every Monday morning, a training coordinator wants to remind people about upcoming sessions. Session details are in Excel. People whose training is marked Completed should not get a reminder.

1. Choose a cloud-flow type.
2. State the start time and time zone.
3. List three pieces of information the flow needs from Excel.
4. Write one rule that decides whether a reminder is sent.

5. Describe a test with one person who should get a reminder and one who shouldn't.

A good plan uses a scheduled flow with a local time zone, needs values like email address, session date, course title and status, sends only for upcoming incomplete training, and tests both an included and an excluded record.

Tips and Troubleshooting

What you notice	What to do
A different Microsoft account opens automatically	Check the account shown on the page. Use the Edge profile meant for training
You can't open Power Automate	Check the address and connection, then ask the trainer to verify access. Note the error message
The environment name is unfamiliar	Compare it with the class instructions before creating anything
Create isn't visible	Expand the left navigation. If it's still missing, ask the trainer to check access
The page looks different from the screenshot	Look for the named command. Layouts vary
A trial or upgrade prompt appears	Check what you selected. The core lessons use the Free baseline
You can't decide which flow type to use	Finish the sentence "This process should begin when..." and match it to a type

Lesson Summary

A trigger starts a flow, actions perform the steps, and data moves between them. Conditions choose a path and loops repeat work for each item. You can now tell automated, instant and scheduled flows apart, find the main Power Automate pages, and describe an automation before you build it.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner.

Automation planning template (section 1.6)

Paste this into your notes and fill in each line before you build a flow. You use the same questions for the independent practice at the end of the chapter.

```
What task repeats?  
What starts it?  
What information is needed?  
What rule applies?  
Where should the output go?  
How will I test it?
```

Power Automate address (section 1.5)

```
https://make.powerautomate.com
```

02

CHAPTER 2

Prepare Your Training Workspace

Every flow in this course reads from or writes to something: a folder, a table, a mailbox, a form, a Teams chat. Set those up first and the later build steps become predictable.

Copy-paste values: the Excel headers, table name and email subject are on the copy-paste page.

Estimated time: 90 minutes

Your result: A training folder, an Excel workbook with a table, and safe test data for the first two build exercises.

What You Will Learn

- Create a dedicated training folder structure in OneDrive for Business
- Build an Excel workbook and format its data as a named table
- Prepare a safe test email with sample attachments
- Confirm that Forms, Teams and your Power Automate environment are ready

Note: Use your own training account and the environment your trainer names. Never put real participant information in the practice resources. Use fictional names and test addresses only.

Forms and Teams preparation in this chapter are trainer-led readiness checks, because those destinations depend on your organisation's policy.

2.1 Create the Training Folders in OneDrive

1. Open OneDrive for Business and select **My files**.
2. Select **Create or upload**, then **Folder**.
3. Name the folder `Power Automate Training` and select **Create**.
4. Open the new folder.
5. Inside it, create two subfolders: `Attachments` and `Reports`.

Checkpoint: You can see `Attachments` and `Reports` inside `Power Automate Training`. The Chapter 3 flow writes to Attachments. Summary outputs can go to Reports.

2.2 Create the Excel Workbook

1. Inside `Power Automate Training`, select **Create or upload**, then **Excel workbook**. Check the breadcrumb first: the workbook is created in whichever folder you're viewing.
2. When Excel for the web opens, select the file name in the title bar and rename it `TrainingRegister`. Excel adds `.xlsx` for you.
3. In Sheet1, enter these headers in row 1:

TrainingID	ParticipantName	Email	CourseTitle	SessionDate	Status
------------	-----------------	-------	-------------	-------------	--------

4. Enter six fictional rows. Include at least two upcoming sessions, one completed session, and one session outside the training week. Use the trainer's approved test addresses in the Email column.
5. Select any cell in the range, choose **Insert > Table**, confirm **My table has headers** is ticked, and select **OK**. Then on the **Table Design** tab, replace the default table name with `tblTraining` and press Enter.

Checkpoint: The workbook is named `TrainingRegister.xlsx`, the data is a formatted table named `tblTraining`, and the table has six fictional records.

Tip: Short on time? Your trainer may tell you to use the ready-made [TrainingRegister.xlsx](#) in this folder instead. It already has the `tblTraining` table and six fictional records. Upload it to `Power Automate Training` in OneDrive. Only do this if your trainer says so.

If Power Automate can't find the table later, check three things: the workbook is in OneDrive for Business, the range is formatted as a table, and the table name is exactly `tblTraining`.

2.3 Prepare a Test Email

1. Open Outlook on the web with the training account.
2. Create a message to your own training mailbox, the one you connect to Power Automate in Chapter 3. Use the subject `[PA TRAINING] Attachment test`.
3. Attach `CourseOutline.pdf` and `TrainerPhoto.jpg` from the [test-files](#) folder in this chapter. **Don't send it yet.** Close the message so it's saved in Drafts.

Important: Don't attach real course materials or any file whose name contains a person's name. The filename shows up in run history and in your OneDrive folder.

The Chapter 3 flow only reacts to new messages, so you send this draft during the test in section 3.6.

2.4 Confirm Forms Readiness

Open Microsoft Forms and confirm the training account can create a form. You build the **Training Request** form in Chapter 6, right before the flow that uses it. It will ask for Requester name, Course, Preferred date and Business reason, with fictional answers only.

2.5 Confirm the Teams Training Destination

Ask the trainer for the team and channel, or the training chat, and write its name down. The trainer confirms posting access before the Teams exercises: the optional extension in Chapter 4 and the approval messages in Chapter 6.

The Teams connector is standard, but your organisation may restrict posting or require the Workflows app. If you can't post a test message, tell the trainer before building the summary flow.

2.6 Confirm the Power Automate Environment

1. Return to Power Automate.
2. Check the environment selector near the top of the page.
3. Confirm it matches the environment your trainer gave you.

Key point: OneDrive files, Excel workbooks, Forms and Teams destinations aren't packaged automatically when a solution is moved. Keep a note of where each resource lives. You'll need it in Chapter 9.

Independent Practice

Find the row in `tblTraining` whose Status is Completed. Which later reminder test should exclude it, and why? Then write a second test email without the `[PA TRAINING]` marker, save it as a draft, and predict whether the attachment flow should process it once it's sent.

Lesson Summary

You created a dedicated OneDrive training area with its folders and Excel table, prepared safe inputs for the attachment and summary exercises, confirmed the environment, and noted where the Forms and Teams work will happen. Next you build the flow that saves qualifying email attachments.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Anything marked **text** is typed or pasted straight into a field.

Folder names (text, section 2.1)

Main folder:

Power Automate Training

Subfolders:

Attachments

Reports

Workbook name (text, section 2.2)

TrainingRegister

Excel column headers (section 2.2)

Click cell A1 in Sheet1, then paste. Each header lands in its own column.

TrainingID	ParticipantName	Email	CourseTitle	SessionDate	Status
------------	-----------------	-------	-------------	-------------	--------

Table name (text, section 2.2)

tblTraining

Test email subject (text, section 2.3)

[PA TRAINING] Attachment test

03

CHAPTER 3

Save Email Attachments Automatically

Your first build: an automated flow that watches for training emails and saves only the PDF attachments to OneDrive.

Copy-paste values: the flow name, subject filter and expressions are on the copy-paste page.

Estimated time: 120 minutes

Your result: An automated cloud flow that examines new training emails and saves only PDF attachments to OneDrive for Business.

Connectors: Office 365 Outlook and OneDrive for Business, both standard.

What You Will Learn

- Create an automated cloud flow from a business requirement
- Configure an Outlook trigger with a subject filter
- Process several attachments with **Apply to each**
- Use a condition to select PDF files
- Create a file in a OneDrive folder
- Test a matching and a non-matching case, then inspect run history

Before You Begin

Complete Chapter 2. Check that the `Attachments` folder exists and the test email from section 2.3 is saved in Drafts. That draft uses the subject `[PA TRAINING] Attachment test` and has two fictional attachments from the Chapter 2 `test-files` folder:

File	Purpose
CourseOutline.pdf	The attachment that should be saved
TrainerPhoto.jpg	The attachment that should be skipped
SessionNotes.docx	Used only in the independent practice

Note: OneDrive never silently overwrites a file. If you run the test twice with the same attachment, Create file either fails with a duplicate-name error or OneDrive saves CourseOutline (1).pdf. Both are correct behaviour. Delete the file from Attachments before a repeat test, or see section 3.8.

The rule: when a new email arrives with [PA TRAINING] in the subject, examine each attachment. If its name ends in .pdf, create a file in the Attachments folder.

3.1 Create the Automated Cloud Flow

1. Open Power Automate and confirm the correct environment.
2. Select **Create**, then **Automated cloud flow**.
3. Name the flow PA - Save training PDF attachments.
4. Search for and select the trigger **When a new email arrives (V3)** from **Office 365 Outlook**.
5. Select **Create**. The designer opens with only the trigger on the canvas.

3.2 Configure the Trigger

1. Set **Include Attachments** to **Yes**.
2. Set **Folder** to **Inbox**.
3. In **Subject Filter**, enter [PA TRAINING].
4. Set **Only with Attachments** to **Yes**. If a field is hidden, use **Show all** under Advanced parameters.
5. Select **Save**.

Checkpoint: The trigger saves without a red validation message. The flow only starts for *new* matching messages. It doesn't go back and process mail already in the inbox.

3.3 Add Apply to Each

1. Select the + insertion button.
2. Search for **Apply to each** and select the Control action.
3. In **Select an output from previous steps**, choose the trigger's **Attachments** dynamic content.

If the Attachments token isn't offered, go back to the trigger and confirm Include Attachments is set to Yes.

3.4 Add the PDF Condition

1. Inside the loop, select **Add an action**.
2. Search for **Condition** and choose the Control condition.
3. In the left value, select **Attachments Name**. If the token picker is awkward, type /, choose **Insert expression**, and enter `item()?['name']`. That reads the current attachment in the loop.
4. Set the comparison to **ends with**.

5. Enter `.pdf` in the right value.
6. In the **True** branch, select **Add an action**.

3.5 Create the OneDrive File

1. Search for **OneDrive for Business**, then pick **Create file** from that connector's group.

Tip: Search by connector name, not action name. Several connectors have an action called Create file, and it's easy to grab the wrong one.

2. In **Folder Path**, select `Attachments` under `Power Automate Training`.
3. In **File name**, select the current attachment's **Name**.
4. In **File content**, select the current attachment's **Content**, or use the expression `item()?['contentBytes']`. After you reopen the designer these may display as `name` and `contentBytes`. Same properties.
5. Leave the **False** branch empty. That's the deliberate decision to skip non-PDF files.
6. Check that Create file sits inside True and that False has no actions.
7. Select **Save**, open **Flow checker**, and confirm 0 errors and 0 warnings.

3.6 Test a Positive Case

1. Select **Test**, choose **Manually**, then **Test**. (Automatically is greyed out on a new flow because it replays a previous run.)
2. Wait for the banner asking you to send a new email. The flow is now listening.
3. Open the draft from section 2.3 and send it.
4. When the run finishes, open it. The loop counter reads **2 of 2**, and Create file succeeded.
5. Open the Attachments folder in OneDrive. The PDF is there. No image file was created.

3.7 Test a Negative Case

You don't need another email. The mixed test already produced both outcomes.

1. Open the same run.
2. In Apply to each, switch the iteration counter from 2 to 1.
3. Select **Condition** and read **expressionResult**. It's `false`, Create file shows a skip marker, and the False branch reports *No Actions*.
4. Check the Attachments folder again. Only the PDF is there.

Key point: A run can succeed while deliberately producing nothing. *Succeeded* means the process completed, not that every branch created an output.

Optional: send `[PA TRAINING] Image-only test` with only `TrainerPhoto.jpg` attached. The run succeeds and creates nothing.

3.8 Decide What Happens to Duplicate Names

Your flow uses the attachment's own name, so the same attachment arriving twice targets the same path. Pick a policy on purpose:

Policy	How	When it suits
Keep one copy	Delete the file from Attachments before re-testing	Classroom testing
Let OneDrive number them	Change nothing	Quick repeat tests where clutter doesn't matter
Make every name unique	Put a timestamp expression in File name	A real process where two senders may use the same file name

The timestamp expression is on the copy-paste page. It produces names like `20260913-094512-CourseOutline.pdf`. `utcNow()` returns UTC, not local time. Chapter 5 covers time zone conversion.

Independent Practice

Change the condition to save `.docx` attachments instead of PDFs. Send one test email with both `SessionNotes.docx` and `CourseOutline.pdf`. Predict which file appears, run it, then restore the `.pdf` rule.

Tips and Troubleshooting

Problem	Check
The flow doesn't run	The message must be new, contain <code>[PA TRAINING]</code> in the subject, and reach the mailbox used by the Outlook connection
Attachments token is missing	Turn on Include Attachments in the trigger, save, and reopen the loop input
The loop runs but creates nothing	Inspect the condition. Left value is the attachment name, right value is <code>.pdf</code> . The comparison isn't case-sensitive
Create file fails	OneDrive connection, folder path, file name and file content token
Duplicate file error	Use a unique test file name or add the timestamp expression
The JPG is saved	Create file is outside the condition. Move it into True

Lesson Summary

This flow combines an event trigger, a loop over attachments, a condition for the business rule and a OneDrive action for the result. You tested both a file-producing path and a successful no-output path. Next, an instant flow reads the Excel table and sends a summary.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Paste expressions into the **Expression** tab of the dynamic content panel (type `/` in a field, then choose **Insert expression**).

Anything marked **text** is typed or pasted straight into a field, not the expression editor.

Flow name (text, section 3.1)

```
PA - Save training PDF attachments
```

Trigger subject filter (text, section 3.2)

```
[PA TRAINING]
```

Condition left value: current attachment name (section 3.4)

```
item()?['name']
```

Condition right value (text, section 3.4)

```
.pdf
```

Create file content (section 3.5)

```
item()?['contentBytes']
```

Unique file name with a timestamp (section 3.8)

```
concat(formatDateTime(utcNow(),'yyyyMMdd-HH:mm:ss'), '-', item()?['name'])
```

Optional image-only test subject (text, section 3.7)

```
[PA TRAINING] Image-only test
```

04

CHAPTER 4

Create and Distribute an Excel Summary

The coordinator wants a readable list of training sessions on demand, without copying cells into an email by hand. An instant flow reads the register and sends one summary table.

Copy-paste values: the flow name, Select mappings and subject are on the copy-paste page.

Estimated time: 120 minutes

Your result: An instant cloud flow that reads the training register, picks useful columns and produces one summary email. A Teams update is an optional extension.

What You Will Learn

- Start a flow on demand with an instant trigger
- Read rows from an Excel table with **List rows present in a table**
- Choose report columns with the **Select** action
- Turn an array into one HTML table and send it in a single email
- Inspect run history to confirm what each action produced

How the Flow Works

Manually trigger a flow > List rows present in a table > Select > Create HTML table > Send an email (V2)

The Excel action returns an array of records. Select shapes each record into the columns readers need. Create HTML table turns the whole array into one table. The email action runs once, after the table is complete.

Before You Begin

You need `Power Automate Training/TrainingRegister.xlsx` in OneDrive for Business with a formatted table named `tblTraining`:

Column	Contents	Used in this chapter for
--------	----------	--------------------------

TrainingID	TR-001 to TR-006	The independent practice
ParticipantName	Fictional names	A report column
Email	Placeholder addresses	Read but deliberately left out of the report
CourseTitle	Fictional course names	A report column
SessionDate	Real Excel dates	A report column, and date formatting in 4.8
Status	Registered or Completed	A report column

Note: Addresses like learner1@example.com are just data in this chapter. example.com is reserved for documentation and can't receive mail. The summary goes to your own training mailbox, not to the people in the table.

Tip: Before each run, finish any cell edit and let Excel save. The connector reads the saved cloud file, so an open edit gives you a stale report.

4.1 Confirm the Source Table

1. Open TrainingRegister.xlsx from the training folder. You built it in section 2.2, so you're only checking it here.
2. Check the header row. All six columns must be present and spelled exactly as above, because the expressions refer to them by name.
3. Select a cell in the data and confirm the table name is exactly tblTraining.
4. Confirm SessionDate holds real dates, not text, and Status uses only Registered or Completed.

Key point: A worksheet name and a table name are different things. The Excel connector needs a formatted table. If the Table dropdown is empty later, this is almost always why.

SessionDate may come back from the connector as a serial number such as 46286. Section 4.3 sets the date format to prevent that.

4.2 Create an Instant Cloud Flow

1. Confirm the training environment and select **Create > Instant cloud flow**.
2. Name the flow PA - Training register summary and choose **Manually trigger a flow**.
3. Select **Create**.

This trigger starts only when you run it. A new email or an edited spreadsheet won't start it.

4.3 Read the Excel Records

1. Add **Excel Online (Business) > List rows present in a table**.

2. Set **Location** to **OneDrive for Business**. (The list also shows every group and SharePoint site you belong to.)
3. Set **Document Library** to **OneDrive**. The other two libraries, KARuntime and PersonalCacheLibrary, are internal and won't have your workbook.
4. In **File**, browse to `Power Automate Training` and pick the workbook. Then choose `tblTraining` in **Table**.
5. Under advanced parameters, set **DateTime Format** to **ISO 8601** if it's available.

If Table shows *No items*, stop. The workbook has no formatted table, and reselecting the file won't fix it. Go back to section 2.2.

4.4 Choose the Report Columns

1. Add **Data Operation > Select** after the Excel action.
2. In **From**, insert the Excel action's **value** output. That's the array of all returned rows.
3. Add these key and value pairs:

Report heading	Value expression
Participant	<code>item()?['ParticipantName']</code>
Course	<code>item()?['CourseTitle']</code>
Session date	<code>item()?['SessionDate']</code>
Status	<code>item()?['Status']</code>

Readers see four useful columns. Email and the internal row metadata are left out, and Select handles every row without you adding an Apply to each.

4.5 Build One HTML Table

1. Add **Data Operation > Create HTML table** after Select.
2. In **From**, select **Output** from Select. Leave columns on automatic, since Select already defines the headings.

Checkpoint: If an Apply to each appears around this action or the email, you picked a single row field instead of the whole array.

4.6 Send the Coordinator Summary

1. Add **Office 365 Outlook > Send an email (V2)** after Create HTML table.
2. Set **To** to your own training mailbox and **Subject** to `[PA TRAINING] Training register summary`.
3. In the body, type a short introduction and insert **Output** from Create HTML table below it. Switch the editor to HTML/code mode if needed so the table renders instead of showing raw tags.
4. Select **Save**, run **Flow checker**, and fix anything it reports.

4.7 Test and Inspect

1. Select **Test > Manually > Test**. This sends a real email, so follow your trainer's instruction before selecting **Run flow**.
2. Open the completed run. Every action has a green check, and Send an email (V2) ran once.
3. Check the Excel output returned six records. Open **Show raw outputs** for Select and confirm the six records have real values, not formula text.
4. Open the email. You should see four headings, six data rows, and one message.

Each manual run sends one current report. Earlier emails stay as historical snapshots.

4.8 Optional Formatting Practice

Start from the working report. In the HTML/code editor, add a short introduction and modest styling: readable font, strong contrast, a header row and decent cell padding. Email clients treat CSS differently, so check the received message after each change.

The row-count and date-format expressions are on the copy-paste page. Check the Excel output before formatting dates, and remember empty dates need a blank-value check before `formatDateTime`.

4.9 Optional Teams Update

Trainer-led only, once a permitted destination is confirmed. After the email, add **Microsoft Teams > Post message in a chat or channel**, choose the approved posting identity and destination, and post a short record count plus a OneDrive link to the workbook. Pasting a link doesn't grant access, so check who can open it. Keep the Teams message short rather than reusing the email's HTML table.

Independent Practice

Add Training ID to the report without adding Email. Predict where the change belongs, make it, and verify the received message. Then explain why a six-record table should still produce only one email.

Troubleshooting

Symptom	What to check
No Excel table is listed	Workbook location, saved table, table name and connector account
One email arrives for every row	Move the email outside any loop and use the Create HTML table output
The report includes internal columns	Use Select to define headings and values
The email shows HTML tags	The body editor mode and the selected output token
A date appears as a number	The DateTime Format setting and the actual run output

The report is stale	Finish the Excel edit, wait for save, and run again
---------------------	---

Teams posting fails	Destination membership, posting identity, policy and Workflows app availability
---------------------	---

Lesson Summary

An instant trigger lets the coordinator decide when to run. Excel supplies the records, Select defines the report columns, and Create HTML table produces one report. Next, a recurrence trigger starts a reminder process on a schedule.

For reference, see the [Excel Online \(Business\) connector](#) and [data operations in Power Automate](#).

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Paste expressions into the **Expression** tab of the dynamic content panel (type `/` in a field, then choose **Insert expression**).

Anything marked **text** is typed or pasted straight into a field, not the expression editor.

Flow name (text, section 4.2)

```
PA - Training register summary
```

Select mapping values (section 4.4)

Participant:

```
item()?['ParticipantName']
```

Course:

```
item()?['CourseTitle']
```

Session date:

```
item()?['SessionDate']
```

Status:

```
item()?['Status']
```

Email subject (text, section 4.6)

```
[PA TRAINING] Training register summary
```

Optional row count (section 4.8)

Works only if the Select action still has its default name.

```
length(body('Select'))
```

Optional date format (section 4.8)

Use this in Select instead of the plain Session date value, only after checking the run output.

```
formatDateTime(item()?['SessionDate'],'dd MMM yyyy')
```

05

CHAPTER 5

Send Weekly Training Reminders

Every week the coordinator opens the register and writes the same reminder for each participant. A scheduled flow can do that at a predictable local time.

Copy-paste values: the flow name and email expressions are on the copy-paste page.

Estimated time: 75 minutes

Your result: A scheduled cloud flow that runs every Monday at 9:00 AM Singapore time, reads the training register, and prepares one reminder email for each row.

What You Will Learn

- Schedule a flow with the **Recurrence** trigger and a local time zone
- Convert a UTC timestamp into readable local time
- Loop through Excel rows with **Apply to each**
- Build a personalised email from the values in the current row

How the Flow Works

Recurrence > Current time > Convert time zone > List rows present in a table > Apply to each > Send an email (V2)

Recurrence controls when the flow starts. Current time supplies a UTC timestamp, and Convert time zone turns it into readable Singapore time for the message. The Excel action returns an array of rows, and Apply to each processes them one at a time.

Before You Begin

You need `Power Automate Training/TrainingRegister.xlsx` with the `tblTraining` table. This chapter uses every column: ParticipantName for the greeting, Email as the recipient, CourseTitle in the subject and body, SessionDate and Status in the body.

Important: The sample register uses addresses like `learner1@example.com`. They can't receive mail, which is what you want while learning. Replace them with trainer-approved test addresses only when the trainer authorises a live test.

5.1 Confirm the Training Register

Open `TrainingRegister.xlsx`, confirm all six columns are present, and check the data is a table named `tblTraining`. Save the workbook and finish any active cell edit.

5.2 Create the Weekly Schedule

1. Select **Create > Scheduled cloud flow**. Name it `PA - Weekly training reminders`, repeat every **1 Week**, select **Monday**, and choose **9:00 AM**.
2. Open **Recurrence** and confirm these values:

Setting	Value
Interval	1
Frequency	Week
Time zone	(UTC+08:00) Kuala Lumpur, Singapore
On these days	Monday
At these hours	9
At these minutes	0

Read the preview. It should say the flow runs at 9:00 on Monday every week. A weekly frequency without a day and local time is incomplete for this scenario.

5.3 Produce a Readable Local Timestamp

1. Add **Date Time > Current time** below Recurrence. It needs no parameters.
2. Add **Date Time > Convert time zone** and set:

Setting	Value
Base time	Current time from the previous action
Source time zone	(UTC) Coordinated Universal Time
Destination time zone	(UTC+08:00) Kuala Lumpur, Singapore
Format string	Full date/time pattern (short time)

Key point: The two time zone settings do different jobs. Recurrence decides *when the flow starts*. Convert time zone only changes a timestamp shown in the message. It doesn't move the schedule.

5.4 Read the Excel Rows

Add **Excel Online (Business) > List rows present in a table:**

Setting	Value
Location	OneDrive for Business
Document Library	OneDrive
File	/Power Automate Training/TrainingRegister.xlsx
Table	tblTraining

If the table list is empty, save the workbook, confirm the range is a formatted table named `tblTraining`, and reselect the file.

5.5 Process One Row at a Time

Add **Control > Apply to each**. In **Select an output from previous steps**, insert **value** from List rows present in a table. The expression version is on the copy-paste page.

Checkpoint: Apply to each needs the Excel **value** array. A single field such as Email or CourseTitle isn't the array.

5.6 Prepare the Reminder Email

Inside Apply to each, add **Office 365 Outlook > Send an email (V2)** and fill the three required fields from the current row:

Field	Value
To	<code>item()?['Email']</code>
Subject	The subject expression from the copy-paste page
Body	The body expression from the copy-paste page

The body greets the participant by name and lists the course, session date, status and the Singapore-time timestamp.

Checkpoint: The email action belongs inside Apply to each. In this version it prepares one message for every row, including Completed ones. Chapter 7 adds the filter.

5.7 Save and Check the Flow

1. Select **Save** and open **Flow checker**. Fix any error before testing.
2. Close Flow checker and confirm the full sequence on the canvas: five top-level steps, with one email action inside Apply to each.

Trainer-Controlled Test

Important: Don't select Test while the workbook still has `example.com` placeholders. A test sends one email per row, so six messages with the sample data. The trainer confirms the recipients and authorises the run immediately before you select Run flow.

After an authorised test, check the run history:

- Recurrence, Current time, Convert time zone and List rows present in a table succeeded
- Apply to each shows the expected number of iterations
- Each Send an email (V2) used that row's Email, CourseTitle, SessionDate and Status
- The reminder timestamp is in Singapore time

Independent Practice

Change the scheduled hour to a value your trainer gives you, read the Recurrence preview, then restore 9:00. Explain why changing Convert time zone wouldn't change when the flow starts.

Troubleshooting

Symptom	What to check
The flow starts at the wrong local time	Recurrence time zone, day, hour, minute and preview
The message timestamp is eight hours early	Convert time zone source and destination
No Excel rows are returned	OneDrive location, workbook save state, formatted table and table name
Apply to each is empty	Use the Excel action's value array
A recipient or course is blank	Header spelling in the workbook and the current-row expressions
Completed records get reminders	Expected in this chapter. Chapter 7 adds the filter
Real messages could be sent	Stop, check every Email value, and get trainer authorisation

Lesson Summary

A scheduled trigger starts the flow at a predictable local time. Date Time actions produce a readable timestamp, Excel supplies the records, and Apply to each hands the current row to the email action. Next you add a human approval decision, and Chapter 7 extends this flow with filtering and a coordinator summary.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Paste expressions into the **Expression** tab of the dynamic content panel (type `/` in a field, then choose **Insert expression**).

Anything marked **text** is typed or pasted straight into a field, not the expression editor.

Flow name (text, section 5.2)

```
PA - Weekly training reminders
```

Apply to each input (section 5.5)

```
outputs('List_rows_present_in_a_table')?['body/value']
```

Email To (section 5.6)

```
item()?['Email']
```

Email Subject (section 5.6)

```
concat('[PA TRAINING] Reminder: ', item()?['CourseTitle'])
```

Email Body (section 5.6)

```
concat('<p>Hello ', item()?['ParticipantName'], ',</p><p>This is your weekly training  
reminder.</p><p><strong>Course:</strong> ', item()?['CourseTitle'], '<br><strong>Session date:  
</strong> ', item()?['SessionDate'], '<br><strong>Status:</strong> ', item()?['Status'], '<br>  
<strong>Reminder generated:</strong> ', body('Convert_time_zone'), '</p><p>Please contact the  
training coordinator if your plans change.</p>')
```

06

CHAPTER 6

Process a Training Approval

A training request needs a person to say yes or no before anything is confirmed. This flow collects the request with Microsoft Forms, asks an approver to decide, and posts a different Teams message for each outcome.

Copy-paste values: the flow name, form text, approval title and messages are on the copy-paste page.

Estimated time: 75 minutes

Your result: An automated cloud flow that reads a Forms response, asks a person to approve or reject it, and prepares a different Teams message for each outcome.

What You Will Learn

- Start a flow when a Microsoft Forms response is submitted
- Retrieve the answers with **Get response details**
- Request a decision with **Start and wait for an approval**
- Branch on the approval outcome and post a different Teams message for each result

How the Flow Works

When a new response is submitted > Get response details > Start and wait for an approval > Condition > Approved or rejected Teams message

The Forms trigger only supplies a response ID. Get response details uses that ID to fetch the answers. The approval action pauses the flow until someone decides. The Condition reads the approval **Outcome** and sends the flow down True or False.

Before You Begin

You need Microsoft Forms, Approvals and Microsoft Teams through your work or school account. Your trainer names an approved test approver and Teams destination before any live test.

Important: The screenshots use `approver@example.com`, a documentation placeholder. Replace it with a trainer-approved account before testing. Don't submit the form or run the flow while the placeholder is still there.

6.1 Prepare the Request Form

In Microsoft Forms, create a form named **Training Request** with the description from the copy-paste page. Add four required questions:

Question	Type	Options or setting
Requester name	Text	Single line
Course	Choice	Power Automate Basics; Approval Workflows; Excel Reporting
Preferred date	Date	Required
Business reason	Text	Long answer

Checkpoint: All four questions show the red required marker. A missing answer gives the approver an incomplete card.

6.2 Create the Automated Flow

Select **Create > Automated cloud flow**. Name it `PA - Training request approval`, search for **Forms**, select **When a new response is submitted**, and select **Create**.

6.3 Select the Form

Open the trigger and set **Form Id** to **Training Request**.

If the form isn't listed, check that you own or can access it, refresh the list, and confirm the Forms connection uses the right account.

6.4 Retrieve the Answers

Add **Microsoft Forms > Get response details**. Set **Form Id** to Training Request, and in **Response Id** insert **Response Id** from the trigger.

The trigger only says a response exists. Get response details returns the individual answers so later actions can use them.

6.5 Request the Decision

Add **Approvals > Start and wait for an approval**:

Field	Value
Approval type	Approve/Reject - First to respond
Title	[PA TRAINING] Training request
Assigned to	Trainer-approved approver
Details	Requester name, Course, Preferred date and Business reason from Get response details

Lay out Details so each label is followed by its matching value.

Checkpoint: Use the values from **Get response details**, not similarly named values from another action.

6.6 Branch on the Outcome

Add **Control > Condition**. Set the left value to **Outcome** from Start and wait for an approval, choose **is equal to**, and enter `Approve`.

Key point: The text must be exactly `Approve`. A typo or a trailing space sends every approval down the False branch.

6.7 Prepare the Approved Message

In the **True** branch, add **Microsoft Teams > Post message in a chat or channel**:

Field	Value
Post as	Flow bot
Post in	Chat with Flow bot
Recipient	Trainer-approved Teams account
Message	Training request approved for followed by Course from Get response details

6.8 Prepare the Rejected Message

In the **False** branch, add a second **Post message in a chat or channel** with the same Post as, Post in and recipient. Set Message to `Training request rejected for` followed by **Course**.

A production process would also handle outcomes like cancelled or timed-out approvals. This exercise sticks to the two standard Approve and Reject buttons.

6.9 Save and Check the Flow

1. Select **Save** and open **Flow checker**. Fix every error before testing.
2. Confirm the canvas: three steps before the Condition, and one Teams action in each branch.

Trainer-Controlled Test

Testing creates an approval request and may post to Teams. Replace both `approver@example.com` placeholders first and confirm the recipient is ready. Then, with the trainer's go-ahead:

1. Submit one fictional Training Request response.
2. Approve it, and check the True branch and the approved Teams message.
3. Submit a second fictional response.
4. Reject it, and check the False branch and the rejected Teams message.

After each test, open run history and confirm the right branch succeeded while the other was skipped.

Independent Practice

Without running the flow, change the approval title so it also includes the **Course** value. Which action supplies that value, and why can't the trigger supply it on its own?

Troubleshooting

Symptom	What to check
Training Request isn't in Form Id	Form ownership or access, the Forms connection account, and a refreshed list
The approval has no request details	Get response details Form Id, the Response Id token and the Details tokens
The flow waits forever	The assigned approver, the notification, and whether a decision was made
An approval goes down False	The Outcome token, <i>is equal to</i> , and the exact text <code>Approve</code>
Teams can't resolve the recipient	Replace the placeholder with a trainer-approved account
No Teams message appears	The branch taken, Teams connection, Post in setting, recipient and the action result
Flow checker is clean but you can't test	Flow checker checks structure only, not placeholder recipients or policy

Lesson Summary

A Forms trigger starts the flow with a response ID, Get response details turns it into usable answers, Start and wait for an approval records a human decision, and a Condition routes the request to the matching branch. The flow is ready for a trainer-controlled test once approved recipients replace the placeholders.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Everything on this page is **text**, typed or pasted straight into a field.

Form name (text, section 6.1)

```
Training Request
```

Form description (text, section 6.1)

```
Submit a fictional training request for the Power Automate approval exercise.
```

Course choices (text, section 6.1)

```
Power Automate Basics  
Approval Workflows  
Excel Reporting
```

Flow name (text, section 6.2)

```
PA - Training request approval
```

Approval title (text, section 6.5)

```
[PA TRAINING] Training request
```

Condition right value (text, section 6.6)

```
Approve
```

Teams messages (text, sections 6.7 and 6.8)

Type the text, then insert **Course** from Get response details after it.

```
Training request approved for
```

```
Training request rejected for
```

07

CHAPTER 7

Capstone: Weekly Reminders and Summary

Chapter 5 sent a reminder to every row, including people who had already finished. The capstone turns that into a process a coordinator would actually use: skip completed training, limit the window, count what was sent, and handle the week when nothing matches.

Copy-paste values: the filter expression, variable name and summary text are on the copy-paste page.

Estimated time: 90 minutes

Your result: A copy of the weekly reminder flow that selects upcoming incomplete records, counts the reminders, and follows a clear summary or no-records branch.

What You Will Learn

- Filter an array before a loop so only relevant rows are processed
- Normalise text and compare dates inside an expression
- Count processed items with an integer variable
- Branch after a loop, including a week with no matching records

How the Flow Works

Recurrence > Current time > Convert time zone > List rows > Filter array > Initialize variable > Apply to each > Send reminder > Increment variable > Condition > Summary or no-records message

The filter sits before the loop, so Apply to each only sees relevant rows. The counter starts at zero and goes up once per reminder. After the loop, a Condition checks the counter and picks the matching branch.

Before You Begin

Complete Chapter 5. The `tblTraining` table must include Email, CourseTitle, ParticipantName, SessionDate and Status.

Important: The sample register uses `example.com` addresses, which can't receive mail. Keep the capstone turned off and don't test it until the trainer has replaced every recipient with an approved destination.

7.1 Make a Safe Capstone Copy

Open `PA - Weekly training reminders`, select **Save As**, enter `PA - Weekly training capstone`, and save. The copy starts turned off.

Checkpoint: Confirm you're editing the capstone copy, not the Chapter 5 flow, before you change anything else.

7.2 Return Excel Dates in a Consistent Format

Open **List rows present in a table**. Under **Advanced parameters**, add **DateTime Format** and select **ISO 8601**. The filter needs a predictable `SessionDate` value to compare.

7.3 Select Upcoming Incomplete Records

1. Add **Data Operation > Filter array** after **List rows present in a table**.
2. Rename it `Filter upcoming incomplete training`.
3. Set **From** to **value** from **List rows present in a table**.
4. Choose **Edit in advanced mode** and paste the filter expression from the copy-paste page.

What the expression does: `trim()` removes stray spaces and `toLowerCase()` ignores capitalisation, so "Completed", "completed " and "COMPLETED" are all excluded. `ticks()` turns each date into a number so dates compare reliably. Both ends of the window, today and today plus 14 days, are included.

Checkpoint: With the sample data and a current date of 17 September 2026, four rows match: 21, 22, 24 and 28 September. The completed rows don't match, including the one on 23 September.

7.4 Start the Reminder Counter

Add **Variables > Initialize variable** after **Filter array**. Rename it `Initialize reminder count`:

Field	Value
Name	ReminderCount
Type	Integer
Value	0

7.5 Loop Through the Filtered Rows

Open **Apply to each** and replace its input with **Body** from **Filter upcoming incomplete training**.

Key point: Don't use the original Excel value array here. That brings back the Chapter 5 behaviour and includes completed and out-of-window rows.

7.6 Count Each Prepared Reminder

Inside Apply to each, after Send an email (V2), add **Variables > Increment variable**. Select **ReminderCount** and enter `1`.

Keep the increment *after* the email. That way the count only includes reminders whose email step completed.

7.7 Review the Learner Reminder

Confirm Send an email (V2) is still inside Apply to each and still uses the current row: To is `item()?'Email'`, Subject and Body are the Chapter 5 expressions. Nothing changes in the email itself. The loop now just feeds it upcoming incomplete records only.

7.8 Decide Whether a Summary Is Needed

Add **Control > Condition** *after* Apply to each, outside the loop. Rename it `If reminders were prepared :`

Left value	Operator	Right value
ReminderCount	is greater than	0

A positive count goes to True. Zero goes to False.

7.9 Prepare the Coordinator Summary

In the True branch, add **Office 365 Outlook > Send an email (V2)** and rename it `Send coordinator summary :`

Field	Value
To	<code>coordinator@example.com</code>
Subject	<code>[PA TRAINING] Weekly reminder summary</code>
Body	The summary sentence from the copy-paste page, with ReminderCount inserted in the middle

7.10 Handle a Week with No Matching Rows

In the False branch, add **Data Operation > Compose**, rename it `No upcoming records`, and enter the no-records sentence from the copy-paste page. That records a clear outcome without sending an unnecessary email.

7.11 Save and Check the Capstone

1. Select **Save**, open **Flow checker**, and confirm zero errors and zero warnings.
2. Close Flow checker, select **Zoom view to fit**, and check the full sequence: filter before the loop, count inside the loop, decision after the loop.

Trainer-Controlled Test

A test can send one learner reminder per matching row plus, when the count is positive, a coordinator summary. With the sample data on 17 September 2026, expect four rows. The trainer replaces every `example.com` address, saves the workbook, confirms the coordinator address and authorises the run first.

Afterwards, check the run history:

- Filter upcoming incomplete training returned the expected rows
- Apply to each ran once per filtered row
- ReminderCount went up once per completed iteration
- A positive count took the True branch and prepared the coordinator summary
- A zero-count test took the False branch and produced the Compose output

Independent Practice

Without running the flow, change the window from 14 days to 7 by editing only the last date expression. Which sample rows would match on 17 September 2026? Then restore 14 days.

Troubleshooting

Symptom	What to check
A completed row passes the filter	Status spelling, <code>trim()</code> , <code>toLowerCase()</code> and the exact text <code>completed</code>
A date comparison fails	DateTime Format is ISO 8601 and SessionDate holds valid dates
Apply to each still includes every row	Use the Body of Filter upcoming incomplete training, not the Excel value array
ReminderCount stays at zero	The variable is initialised before the loop and incremented inside it
The summary runs before reminders finish	Put the Condition after Apply to each, outside its container
The False branch has an email action	Use Compose for the no-records result
Flow checker is clean but a test is unsafe	Flow checker checks structure. It doesn't approve recipients or stop messages

Lesson Summary

The capstone filters before looping, normalises text inside an expression, compares dates as numbers, keeps an integer counter, and branches on the final count. That turns a broad weekly reminder into a controlled process with a clear result for the coordinator.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Paste expressions into the **Expression** tab of the dynamic content panel (type `/` in a field, then choose **Insert expression**).

Anything marked **text** is typed or pasted straight into a field, not the expression editor.

Flow name for the copy (text, section 7.1)

PA - Weekly training capstone

Filter array action name (text, section 7.3)

Filter upcoming incomplete training

Filter array condition, advanced mode (section 7.3)

```
@and(
  not(equals(toLower(trim(item()?['Status'])), 'completed')),
  greaterOrEquals(ticks(item()?['SessionDate']), ticks(startOfDay(utcNow()))),
  lessOrEquals(ticks(item()?['SessionDate']), ticks(addDays(startOfDay(utcNow()), 14)))
)
```

Initialize variable action name (text, section 7.4)

Initialize reminder count

Variable name (text, section 7.4)

ReminderCount

Apply to each input (section 7.5)

body('Filter_upcoming_incomplete_training')

Condition name (text, section 7.8)

If reminders were prepared

Coordinator summary action name (text, section 7.9)

Send coordinator summary

Coordinator summary subject (text, section 7.9)

[PA TRAINING] Weekly reminder summary

Coordinator summary body (text, section 7.9)

Paste the first part, insert **ReminderCount** from dynamic content, then paste the second part.

The weekly training reminder flow prepared

reminder(s) for upcoming incomplete sessions in the next 14 days.

No-records Compose name (text, section 7.10)

No upcoming records

No-records Compose input (text, section 7.10)

No upcoming incomplete training records were found for the next 14 days.

08

CHAPTER 8

Plan an Automation with Copilot

Copilot in Power Automate can turn a plain-language description into a suggested flow. A suggestion isn't a finished design, though. In this chapter you review what Copilot proposes, correct it precisely, and walk away without creating anything.

Copy-paste values: the three prompts for this chapter are on the copy-paste page.

Estimated time: 30 minutes

Your result: A reviewed and improved flow plan produced with Copilot, without creating or running a cloud flow.

What You Will Learn

- Describe an automation to Copilot with a clear, bounded prompt
- Review an AI-generated flow plan against the business rule
- Refine a suggestion by naming the exact change you need
- Leave the planning experience without creating a flow

The Business Problem

Copilot may leave out a calculation, add an action you don't need, or read a broad instruction differently from what you meant. You ask it to plan a weekly coordinator summary, check every proposed step, and correct it until it matches this intended plan:

Recurrence > List rows present in a table > Filter array > Compose count > Send one coordinator email

Before You Begin

Copilot availability depends on your organisation, environment, region and admin settings. If **Create with Copilot** or **Describe it to design it** isn't available, follow along with the manual's screenshots and answer the review questions without entering a prompt.

Important: Use only fictional details and `coordinator@example.com`. Never put confidential information, personal data, passwords or real recipients into a training prompt.

8.1 Open the Copilot Planning Experience

On the Power Automate home page, select **Create with Copilot**. The Create page may call the same thing **Describe it to design it**.

8.2 Describe the Complete Business Outcome

In **What will your flow do?**, paste prompt 1 from the copy-paste page and select **Submit**.

A strong prompt states the event, the data source, the rule, the output and the boundaries. Even so, Copilot doesn't know your approved recipient, the exact workbook location, your organisation's policies, or how you'll test the result.

8.3 Review the First Suggestion

Read the trigger and every proposed action before you select anything. In the manual's example, the first version suggested Recurrence, List rows present in a table, FilteredRows, Initialise Count and Send an email. It looks plausible, but **Initialise Count** doesn't say how the count is calculated.

Use these questions on any AI-generated flow plan:

Review area	Question
Trigger	Does it start at the intended event, time and time zone?
Data	Does it use the correct connector, file, table, list, form or mailbox?
Logic	Are filtering, looping, conditions and calculations explicit?
Output	Is the destination approved, and could the action contact real people?
Licensing	Are all proposed connectors and actions available to participants?
Testing	Can the plan be tested safely with fictional data and approved destinations?

8.4 Clarify the Filter and Count

In **Add more details to improve the flow**, paste prompt 2 and select **Send**. It names the actions, gives the counting expression, and draws a clear line around who gets email.

8.5 Identify an Unnecessary Action

The second suggestion now has a proper Count and still one email, but it adds **Get file metadata using path** before List rows present in a table. You don't need it: List rows present in a table can select the workbook directly. The extra action is just another dependency.

8.6 Request a Streamlined Design

Paste prompt 3 and select **Send**. It says what to remove, why, and exactly which structure to keep.

8.7 Validate the Final Suggestion

The third suggestion should have exactly these steps:

Order	Proposed step
1	Recurrence
2	List rows present in a table
3	FilteredRows
4	Compose Count
5	Send an email

This only checks the outline. If you continued, you'd still need to configure the Friday schedule and time zone, the workbook and table, the filter and count expressions, the recipient, subject and body, error handling, and test data.

8.8 Exit Without Creating the Flow

Select **Cancel**, not **Keep it and continue**. Power Automate returns to the Create page. This exercise is planning only.

Independent Practice

Write a prompt for a monthly training-capacity report. Include the schedule, data source, inclusion rule, output and one explicit boundary. Swap prompts with another participant and find one ambiguity before asking Copilot for a suggestion.

Troubleshooting

Symptom	What to check
Create with Copilot is missing	Organisation settings, environment, region, account entitlement. Use the manual's example instead
Copilot suggests the wrong trigger	State the exact event or schedule: frequency, day, time and time zone
The plan has a participant email loop	Say only one coordinator summary is needed and explicitly rule out participant emails

The count is unclear	Name the filtered array and ask for <code>Length(body('FilteredRows'))</code>
An unnecessary connector appears	Explain which existing action already provides the data and ask Copilot to remove the extra one
The final outline looks right	Keep reviewing. Fields, expressions, connections, recipients and testing aren't configured yet

Lesson Summary

Copilot can propose a trigger and action sequence quickly, keep versions, and respond to refinements. You stay responsible for the business rule, connector choice, licensing, recipient safety, expressions, configuration and testing. The best result came from reviewing each suggestion and stating the exact correction.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner, then paste into Copilot.

These prompts use fictional details only. Never put confidential information, personal data, passwords or real recipients into a training prompt.

1. Describe the complete business outcome (section 8.2)

Paste into **What will your flow do?** and select **Submit**.

```
Every Friday at 4:00 PM Singapore time, read rows from the tblTraining table in TrainingRegister.xlsx in OneDrive for Business. Keep rows whose Status is Registered and SessionDate is within the next 14 days. Email the training coordinator a summary count. Do not send participant emails.
```

2. Clarify the filter and count (section 8.4)

Paste into **Add more details to improve the flow** and select **Send**.

```
Use Filter array to keep rows where Status is Registered and SessionDate is between today and 14 days from today. Calculate the count with length(body('FilteredRows')). Send one summary email only to coordinator@example.com. Do not add an Apply to each or any participant email.
```

3. Request a streamlined design (section 8.6)

```
Remove Get file metadata using path because List rows present in a table can select the workbook directly. Keep only Recurrence, List rows present in a table, Filter array named FilteredRows, Compose named Count using length(body('FilteredRows')), and one Send an email action to coordinator@example.com.
```

Remember: select **Cancel** at the end (section 8.8). This exercise is planning only.

09

CHAPTER 9

Move Automations with Environments and Solutions

A flow that works in one environment can still be hard to move safely. It depends on connections, tables and other components, and copying them one at a time makes it easy to miss something. Solutions package those pieces together.

Copy-paste values: the solution names are on the copy-paste page.

Estimated time: 60 minutes

Your result: An unmanaged solution package containing the capstone flow and its connection references, plus a safe walkthrough of where an import starts.

What You Will Learn

- Explain what an environment is and why organisations separate development from production
- Describe what a solution can contain
- Explain how Power Automate, Power Apps and Copilot Studio can share solution components
- Create a solution and add an existing cloud flow
- Publish and export an unmanaged solution
- Identify the safe import, connection-repair, verification and testing sequence

The Migration Path

Development environment > Solution > Publish > Export package > Target environment > Import > Repair connections > Verify > Authorised test

The live demonstration stops after opening the import screen. Importing back into the same environment isn't a real migration test, and testing the capstone could send email.

Before You Begin

This is trainer-led unless every participant has an environment with Microsoft Dataverse and permission to create, export and import solutions. Power Automate Free alone doesn't guarantee those features.

The worked example uses **PA - Weekly training capstone** from Chapter 7. Keep it turned off. Don't import or test it unless the trainer supplies a separate target environment, approves the destinations and authorises the run.

Environments in Plain Language

An environment is a separate workspace for apps, flows, data, connections and security. The environment selector at the top of Power Automate shows which one you're in.

Environment	Typical purpose	Practical rule
Development	Build and change automations	Use fictional or approved test data
Test	Validate behaviour before release	Reconnect services and run controlled tests
Production	Run approved business processes	Restrict changes and monitor real operation

Separating these stages stops unfinished changes reaching real users, but only when permissions, connection ownership, data and release steps are controlled too.

Power Automate, Power Apps and Copilot Studio all sit on the Power Platform, so one solution can package the flows, apps, agents, tables, connection references and environment variables behind a single business process. What's available depends on the environment and licences.

9.1 Confirm the Current Environment

Check the environment selector before you create or import anything, and confirm with the trainer that it's the development or training environment. If a flow or solution seems to be missing, check the environment before rebuilding it.

9.2 Create the Solution

Open **Solutions**, select **New solution**, and enter:

Field	Value
Display name	Power Automate Training Migration
Name	PowerAutomateTrainingMigration
Publisher	The trainer-approved publisher
Version	1.0.0.0

Then select **Create**. The display name is the friendly label. The unique name is used internally and normally shouldn't change later.

9.3 to 9.7 Add the Capstone Flow

1. Open the new solution. Its object list is empty: a new solution is just a container.
2. Select **Add existing** and point to **Automation**.
3. Under Automation, select **Cloud flow**.
4. Select **Outside Dataverse** and find `PA - Weekly training capstone`. Flows that aren't in a solution yet are listed here.
5. Select the capstone, then **Add**.

Power Automate may also add **connection references**. A connection reference is a solution component that points to a connection. The target environment still needs a valid connection and an authorised account.

9.8 Verify the Solution Components

Confirm the solution has three objects: the capstone cloud flow and the Excel Online (Business) and Office 365 Outlook connection references.

Key point: Showing up in the object list doesn't mean a dependency is ready. File locations, table names, recipients, connection accounts, environment variables and permissions can all differ in the target environment.

9.9 Review the Solution Before Export

Open **Overview** and check the name, version, package type, publisher and included items.

9.10 to 9.12 Publish and Export

1. Select **Export**. On **Before you export**, select **Publish** so the latest changes are included.
2. Wait for **Published**, then select **Next**.
3. Confirm the version, select **Unmanaged**, and select **Export**. Power Automate suggests the next version number (for example 1.0.0.1), so every exported package has its own version.
4. When the success message appears, select **Download** and store the ZIP in the trainer-approved location. The worked example produced `PowerAutomateTrainingMigration_1_0_0_1.zip`.

An unmanaged package suits this exercise because its components can be edited after import. Managed solutions are normally used for controlled distribution where recipients shouldn't edit the components. That's an organisational decision, so don't switch package types casually in an established process.

Tip: Don't unzip or rename the package before import. Treat it as a release artifact and record who exported it, from which environment, and why.

9.13 and 9.14 Switch Environment and Start the Import

1. Use the environment selector to choose the trainer-approved target environment, and confirm its name before opening **Solutions**.
2. Select **Import solution**, then **Browse** to the exported ZIP. Select **Next** only after confirming the package name and the target environment.

The demonstration stops here. No file is uploaded and nothing is imported. Your trainer may not switch environments at all if there's no second training environment. **Never use production as a stand-in for a training target.**

9.15 Repair Connections and Settings

After the package is selected, Power Platform may ask for connection mappings or environment-specific values. Before import or first use:

1. Map each connection reference to an approved connection in the target environment.
2. Confirm the Excel workbook and `tblTraining` table exist in the intended target location.
3. Review recipients, team and channel selections, form references, schedules, time zones and environment variables.
4. Confirm the importing account and future flow owner have the required permissions.
5. Keep the imported flow turned off until verification is complete.

9.16 Verify and Test Safely

After an authorised import, open the solution and confirm the expected components are there. Run Flow checker, review every connection and setting, and use fictional data and approved destinations for the first test.

Testing is a separate release decision. An imported flow can be structurally valid and still point at the wrong workbook, mailbox, form, team, channel or recipient. Get trainer authorisation right before any test that can send messages or change shared data.

Independent Practice

Create a migration checklist for one of the earlier flows. Name its trigger, connectors, data sources, recipients, environment-specific settings, owner, safe test data, and the evidence you'd need before turning it on in production. Don't export, import or run anything.

Troubleshooting

Symptom	What to check
Solutions is missing or unavailable	Environment type, Dataverse, licence, security role and maker permissions

The flow isn't in the list	Current environment, the Outside Dataverse tab, flow ownership, and whether it's already in another solution
Extra components appear	Connection references and other dependencies may be added automatically. Review each one
Export stays pending	Refresh Solutions, open export history, and confirm publishing finished
Import says the solution already exists	Check you're in the intended target, and whether this is an upgrade or an accidental same-environment import
A connection shows an error after import	Create or pick an approved target connection and remap the connection reference
The flow imports but can't find the workbook	Target OneDrive or SharePoint location, file name, table name, permissions and connector account
The imported flow is off	Leave it off while reviewing. Turn it on only through the approved release process

Lesson Summary

Environments separate workspaces, permissions, data and release stages. Solutions package related Power Platform components so a process can move in a controlled way. A safe migration covers dependency review, publishing, a versioned export, a confirmed target, connection mapping, configuration checks, Flow checker, controlled testing and a deliberate decision to switch the flow on.

HANDS-ON

Hands-on exercises

Hover over a grey box and click the copy icon in its top-right corner. Everything on this page is **text**, typed or pasted straight into a field.

Solution display name (text, section 9.2)

```
Power Automate Training Migration
```

Solution name (text, section 9.2)

```
PowerAutomateTrainingMigration
```

Solution version (text, section 9.2)

```
1.0.0.0
```

Microsoft Power Automate for End Users

This book accompanies the two-day Microsoft Power Automate for End Users course. It follows one training coordinator's routine, from saving email attachments and sending an Excel summary, through weekly reminders and a Forms approval, to a filtered capstone flow, a Copilot-planned design and a solution ready to move. Every chapter pairs short notes with copy-ready names and expressions you can use during the session and back at your desk.